

Intensified liquid-liquid process design for critical metals extraction from e-waste

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ABSTRACT

Critical metals are essential for clean energy technologies but, due to being mainly sourced through mining, the critical metal supply chain is susceptible to geopolitical risks. Electronic waste (e-waste), however, can serve as an alternative “urban mine”, but the recovery at high purities requires complex and resource-intensive processing. This work explores the modeling and optimization-based design for the intensification of liquid-liquid extraction in small channels as a means to recover critical metals from e-waste. Small channels can achieve high mass transfer rates while mitigating the environmental impact. A superstructure-based approach is employed to represent the alternative system configurations, while a plant propagation algorithm is used to optimize the multi-objective problem to recover Neodymium (Nd) and Samarium (Sm). The multi-objective problem aimed to tackle product quality, process economics, and environmental impact. The results demonstrated that optimally designed extraction can support efficient recovery of critical metals; however, decision making support may be essential for the selection of one process design from those identified as optimal in a multi-objective design approach.

Keywords: Superstructure optimization, multi-objective optimization, extraction, critical metals, process intensification

INTRODUCTION

Critical metals are integral to renewable low carbon and hydrogen technologies and essential for the global energy transition since they are necessary to produce solar panels, batteries, wind turbines, electronics and catalysts (Department for Business & Trade, 2025). These metals are mainly sourced through mining; however, only few countries have access to available ores, with China controlling more than 90% of the global rare earth market and South Africa controlling most of the platinum group metals market (*Recycling Rare Earth Metals - Mathys & Squire LLP*, n.d.). With the global demand in critical metals being higher than ever, their availability has become susceptible to geopolitics, thereby threatening the supply chains. Alternatives to mining include the recovery of critical metals from the increasing volumes of electronic waste (e-waste), which can serve as “urban” mines with critical metal concentrations often higher than

in primary ores (Hagelüken, 2012). However, the recovery of metals from e-waste at required purities necessitates multiple processing steps, including solvent extraction and recovery steps.

Metals recovery from waste is traditionally carried out via pyrometallurgical or hydrometallurgical routes. Hydrometallurgical routes are preferred since these do not involve hazardous emissions and offer high recovery and selectivity for a wide range of metals. Hydrometallurgical routes involve the dissolution of the solid waste in acidic solutions followed by a solvent extraction step responsible for selectively separating the metals into solvent phases. Traditionally, dissolution and extraction are carried out in batch systems, and thus are usually characterized by low efficiency, while they are difficult to monitor and digitize due to non-uniformities associated with the batch processing. In lieu of batch dissolution and extraction, intensified flow processing in small channels presents an alternative that can enable the transition to

fully monitored continuous processes, owing to enhanced mass transfer rates that can eliminate flow non-uniformities, increase efficiency, and decrease solvent usage. Throughput in intensified small channel processing can be increased in a modular way by increasing the number of channels and not their volume, thus also facilitating an easier transfer from lab to pilot to industrial settings and ensuring process safety.

In this work, we explore the multi-objective optimal design for the separation of Neodymium (Nd) and Samarium (Sm), primarily used for magnets found in electronics and electric vehicles, using continuous flow liquid-liquid extraction (LLE) in small channels (Garciadiego-Ortega et al., 2020). Mass transfer in the channels is modeled as a set of differential equations. We follow a superstructure-based approach to determine the optimal flowsheet by accounting for multiple extraction steps (Mencarelli et al., 2020). Each step may consist of several stages. The multiple objectives reflect final products quality, process economics and environmental footprint (e.g. solvent usage). The optimization problem is solved employing Fresa, a nature inspired plant propagation optimization algorithm (Fraga, 2021).

This manuscript is structured as follows: first, an overview of the extraction system under consideration is presented. Then, the modeling components, simulation and optimization methods are provided. The proposed system is then used to implement a case study and the results are discussed. Finally, some concluding remarks and future directions are provided.

METHODOLOGY

System under consideration

For this work, we considered the continuous flow LLE system in small channels adapted from Pineda et al., (2021). In the channels, high mass transfer rates can be achieved by inducing segmented flow, that is a repeating pattern of two immiscible liquids moving as elongated drops (plugs) separated by continuous phase slugs. The reader is referred to the work of Garciadiego-Ortega et al. (2020) for a detailed presentation of segmented flow in small channels. This extraction system employs two extraction steps and one scrubbing step. A feed of known concentration and a stream of fresh organic solvent enter the first extraction step, E1. For the organic solvent, we assumed a mixture of 2-ethylhexyl diamide amine in n-dodecane. The organic outlet of E1 is introduced into a scrubbing step, SCR, to further purify the organic stream, whilst the aqueous outlet of E1 is introduced into subsequent extraction step, E2. The mixers in the flowsheet of Fig. 1 are used to combine the two distinct organic and aqueous streams, respectively. Note that the mixers used in this work do not appear in the work of Pineda et al. (2021) since the considered system was used to

optimally detect trivalent Samarium and Neodymium in nuclear waste, while also considering uncertainty. In the work of Pineda et al. (2021), each extraction step was equivalent to a single stage, and thus a single channel. The current work aims to identify optimal designs for the recovery of the two metals employing multi-stage extraction steps. The number of stages is a design variable as explained in a later section of this paper. Fig. 2 depicts a multi-stage configuration where the flow is introduced co-currently in each stage (channel) but counter-currently overall in a given processing step: the feed is introduced in the first stage of a step whilst the solvent is introduced in the last stage. This pseudo-counter-current configuration aids in more efficient separations (Bascone et al., 2019) and thus is preferred to co-current configurations.

Modeling

Modeling solvent extraction in small channels is a problem well addressed in the open literature. Although computational fluid dynamics (CFD) models offer rigor and accuracy (Zhang et al., 2019), they can hinder the solution of the optimization problem due to the computational burden that arises from such models. Instead, we can use a simplified model that consists of two differential mass balances, for the aqueous and organic phases (Pineda et al., 2021). Thus, the differential mass balance for the aqueous phase concentration $C_{a,i,j}$ of the i -th component of the j -th processing step reads:

$$v_{a,j} \frac{\partial}{\partial x} C_{a,i,j}(x) = -K_j [C_{a,i,j}(x) - C_{a,i,j}^*(x)] \text{ for } x = 0, \dots, L_j \quad (1)$$

where $C_{a,i,j}^*$ is the equilibrium concentration given by $C_{a,i,j}^* = C_{o,i,j}/a_i$ with a_i being the distribution coefficient of the i -th component. We used values of $a_{Nd} = 3.7$ and $a_{Sm} = 0.3$, for the extraction and scrubbing steps described above, that come from the work of Suzuki et al. (2017). The mass balance of the organic phase concentration $C_{o,i,j}$ of the i -th component of the j -th step reads:

$$\frac{\partial}{\partial x} C_{o,i,j}(x) = -q_j \frac{\partial}{\partial x} C_{a,i,j}(x) \text{ for } x = 0, \dots, L_j \quad (2)$$

where K_j is the mean overall volumetric mass transfer coefficient, L_j is the channel length, $q_j = \frac{\dot{Q}_{a,j}}{\dot{Q}_{o,j}}$ is the ratio between the volumetric flow rate of the aqueous phase, $\dot{Q}_{a,j}$, and the organic phase, $\dot{Q}_{o,j}$, of each step, $v_{a,j} = \frac{\dot{Q}_{a,j}}{A_j}$ is the superficial velocity of the aqueous phase, and $A_j = \pi D_j^2/4$ is the cross sectional area of each channel in a processing step. To estimate the mean overall volumetric mass transfer coefficient, K_j , a hydrodynamics correlation is used (Kashid et al., 2010):

$$K_j = \left(\frac{\dot{Q}_{a,j} + \dot{Q}_{o,j}}{A_j} \right) \left(\frac{Ca^{-0.09} Re^{-0.09}}{L_j} \right) \left(\frac{D_j}{L_j} \right)^{-0.1} \quad (3)$$

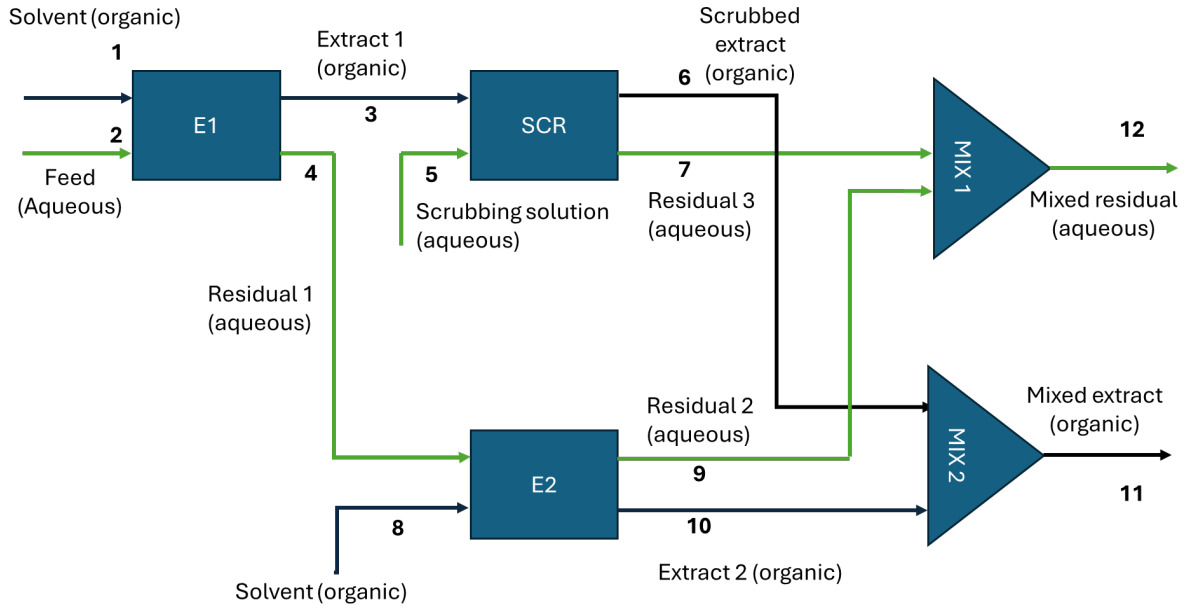


Figure 1. Schematic of the continuous flow LLE system considered in this work.

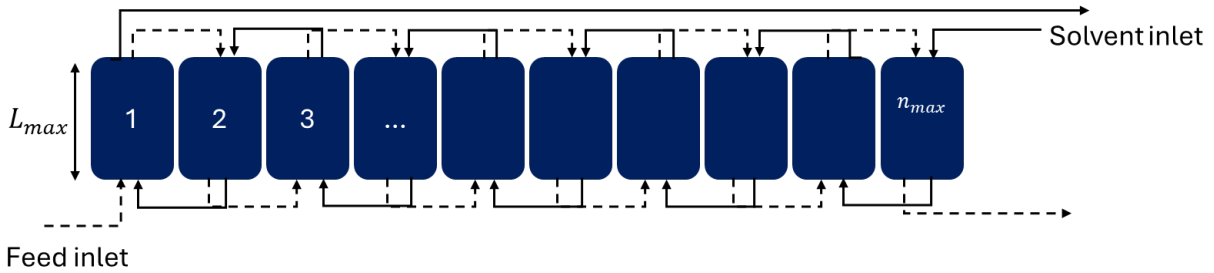


Figure 2. Schematic of the pseudo-countercurrent multi-stage configuration within each step of the system considered in this work.

where Ca and Re are the capillary and Reynolds numbers, respectively, and D_j is the diameter of the channels of each step.

Simulation and optimization

We employ a superstructure-based approach to determine the optimal flowsheet for the separation process (Mencarelli et al., 2020). The process flowsheet involves two distinct extraction steps and a scrubbing step as depicted in Fig. 1. Each extraction step may comprise of several stages (in this work: single-channel stages). The Julia programming language (Bezanson et al., 2017) is used to model the flowsheet as a superstructure of extraction steps (scrubbing included), and in turn, each extraction step as a superstructure of stages. This approach offers good scalability and modularity. The pseudo-countercurrent configuration (see Fig. 2), implemented in Julia, results in multiple recycle streams. The streams have to be torn and converged in order to simulate the superstructure. Therefore, to converge the

streams we have implemented Broyden's method in Julia (Broyden, 1965).

Optimal designs are generated by considering a search space defined by the decision variables vector $\mathbf{x} = [q_j, \xi_j, D]$, where ξ_j is a decision variable that defines both the number of stages n_j as well as the length L_j of the j -th step (i.e., channel). Note that we assume the length stays uniform within a given step, that is all channels share the length, although it can vary across the different processing steps. This assumption aims to facilitate structural simplicity of the flowsheet in order to maintain low capital cost and promote scalability of the plant. In parallel, a distinct channel length variable for each stage would drastically increase the search space and potential solutions, thus leading to an exponential increase in computational time.

Therefore, given $\xi_j \in [0, 1], L_{\max,j}, L_{\min,j}, n_{\max,j}$:

$$n_j \leftarrow \lceil \xi_j n_{\max,j} \rceil \quad (4)$$

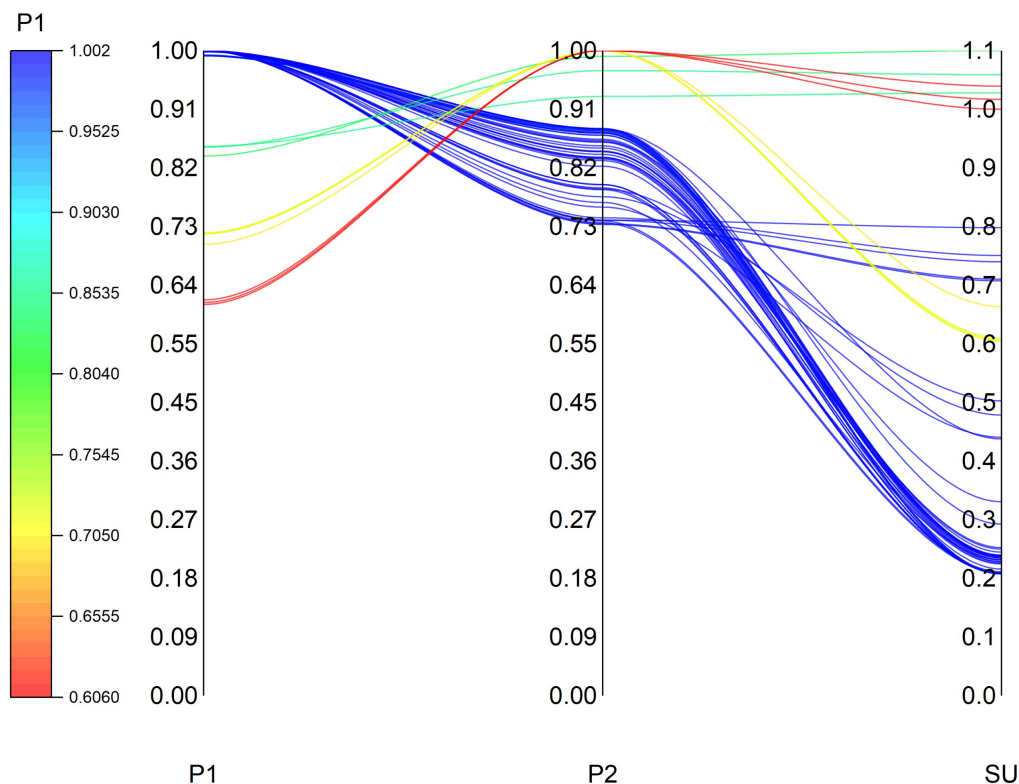


Figure 3. Parallel coordinate plot of the non-dominated solutions obtained by Fresa for the settings in Table 4 to solve the three-objective problem defined by Eqs. (6)–(8). P1 and P2 correspond to the purities of Nd and Sm, respectively, while SU is the organic solvent usage in m³/yr.

$$L_j \leftarrow (\xi_j n_{\max,j} - n_j + 1)(L_{\max,j} - L_{\min,j}) + L_{\min,j} \quad (5)$$

The formulation described in Eqs. 4 and 5 is useful because a single value for ξ_j (not physically meaningful) is passed down to define two physically meaningful design variables, n_j and L_j . This decreases the dimensionality of the problem, thus reducing the potential computational cost. Moreover, the above formulation deals with discrete decisions (e.g., the number of stages per step) by relying solely on continuous variables. This mitigates the combinatorial explosion that would be caused if we employed discrete or binary variables instead. This representation of the design space is effective since it includes all the designs of interest, yet it mitigates any infeasible designs. However, there will be potential designs, that albeit feasible, might be impractical.

We employed Fresa to solve the optimization problem. Fresa is an implementation of a plant propagation algorithm in the Julia programming language (Fraga, 2021). Fresa is a nature-inspired evolutionary algorithm that mimics the propagation of strawberry plants. In fertile soil, strawberry plants send runners to exploit (i.e. refine) the surroundings, while in less fertile soil, fewer but longer runners are sent to explore the *design space* to

find more fertile soil. We refer the reader to original paper Salhi & Fraga (2011) for a detailed description of the plant propagation algorithm.

CASE STUDY AND RESULTS

Neodymium (Nd) and Samarium (Sm) are metals that can be considered as rare earth elements. Applications of these metals include, but are not limited to, magnets for electronic devices, electric vehicles (EVs), and wind turbines. Hence, these metals are pivotal in energy transition. Pineda et al. (2021) have explored optimal design of continuous flow LLE in small channels with the aim to optimally *detect* Nd and Sm radionuclides during the decommissioning of nuclear power plants. In this work, we drew data from the work of Pineda et al. (2021) to develop designs to achieve the best *separation* of Nd and Sm.

As stated above, the main goal of this work is to intensify the separation of critical metals in order to achieve high metals purities and recoveries, while minimizing the environmental impact and operational cost by minimizing the organic solvent usage. The above goals are addressed through the definition of a three-objective

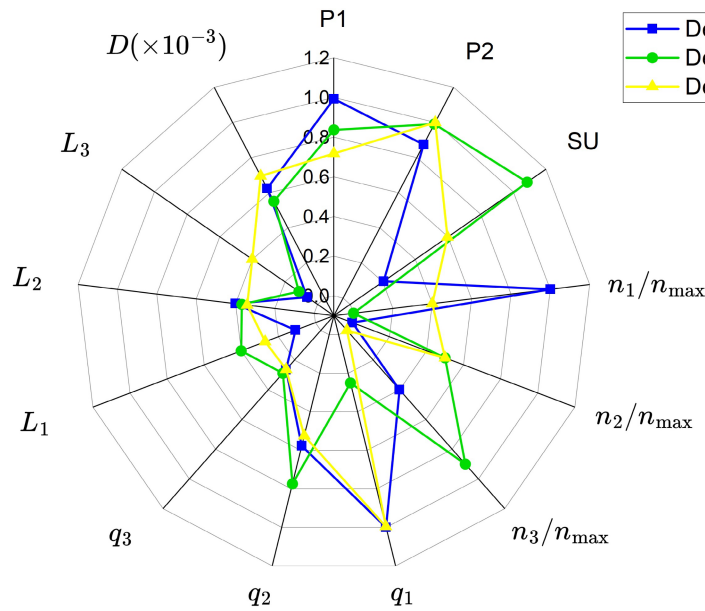


Figure 4. Radar chart of representative designs drawn from the set of non-dominated solutions (objectives and design variables) of the three-objective problem defined by Eqs. (6)-(8).

optimization problem:

$$\max_{\mathbf{x}} f_1(\mathbf{x}) = \frac{C_{Nd,11}}{C_{Nd,11} + C_{Sm,11}} \equiv \text{Purity}_{Nd} \quad (6)$$

$$\max_{\mathbf{x}} f_2(\mathbf{x}) = \frac{C_{Sm,12}}{C_{Nd,12} + C_{Sm,12}} \equiv \text{Purity}_{Sm} \quad (7)$$

$$\min_{\mathbf{x}} f_3(\mathbf{x}) = (\dot{Q}_1 + \dot{Q}_8) \frac{m^3}{s} \times 31,536,000 \frac{s}{yr} \quad (8)$$

Note that while only the purities of the two components are explicitly optimized, the component recoveries are also implicitly optimized, due to the Purity of Nd being associated to the Recovery of Sm and vice versa. To gain better insight into why this happens, the recoveries of the components read:

$$\text{Recovery}_{Nd} \equiv \frac{C_{Nd,11} \dot{Q}_{11}}{C_{feed} \dot{Q}_{feed}} \quad (9)$$

$$\text{Recovery}_{Sm} \equiv \frac{C_{Sm,12} \dot{Q}_{12}}{C_{feed} \dot{Q}_{feed}} \quad (10)$$

From Eqs. (6), (7), (9) and (10), we can deduce that to increase the purity of (for instance) Nd the concentration of Sm in Stream 11 must be reduced. Since there are only two final outlet streams as depicted in Fig. 1 (Streams 11 and 12), reducing the concentration of Sm in Stream 11 translates into concentrating Sm in Stream 12, and thus increasing its recovery. The same stands for the Purity of Sm and Recovery of Nd. Note also that Eq. 8 targets only the organic solvent streams (Streams 1 and 8) and accounts for solvent consumption in a yearly basis operation.

The extraction system consists of four streams of

known concentration. The feed enters E1 at a given metal concentration and known flow rate. A stream of fresh organic solvent also enters E1 whose flow rate is determined by the design variable, q_1 . Stream 8 that contains fresh organic solvent enters a subsequent extraction step, E2, at a flowrate determined by q_2 , whilst the fresh aqueous solvent stream that enters the scrubbing step, Stream 5, enters at a flowrate determined by q_3 . Table 1 summarizes the streams of known concentrations.

Table 1: Streams of known concentrations in the liquid-liquid extraction flowsheet. Flow rate values that depend on the optimal designs are denoted by To Be Defined (TBD).

Stream	Description	Concentration (M)		Flow rate (m ³ /s)
		Nd	Sm	
Feed (2)	Aqueous	0.001	0.001	1.3×10^{-8}
1	Organic	0.0	0.0	TBD
8	Organic	0.0	0.0	TBD
5	Aqueous	0.0	0.0	TBD

The mass balance of Eq. 1 features the mean overall volumetric mass transfer coefficient which is calculated based on Eq. 3. The dimensionless numbers featured in Eq. 3 require values for the physical properties of the mixture that flows through the channel. Thus, we used values for the interfacial tension, the viscosity, and density of the mixture that we drew from the work of Pineda et al. (2021). The values for both the organic and the

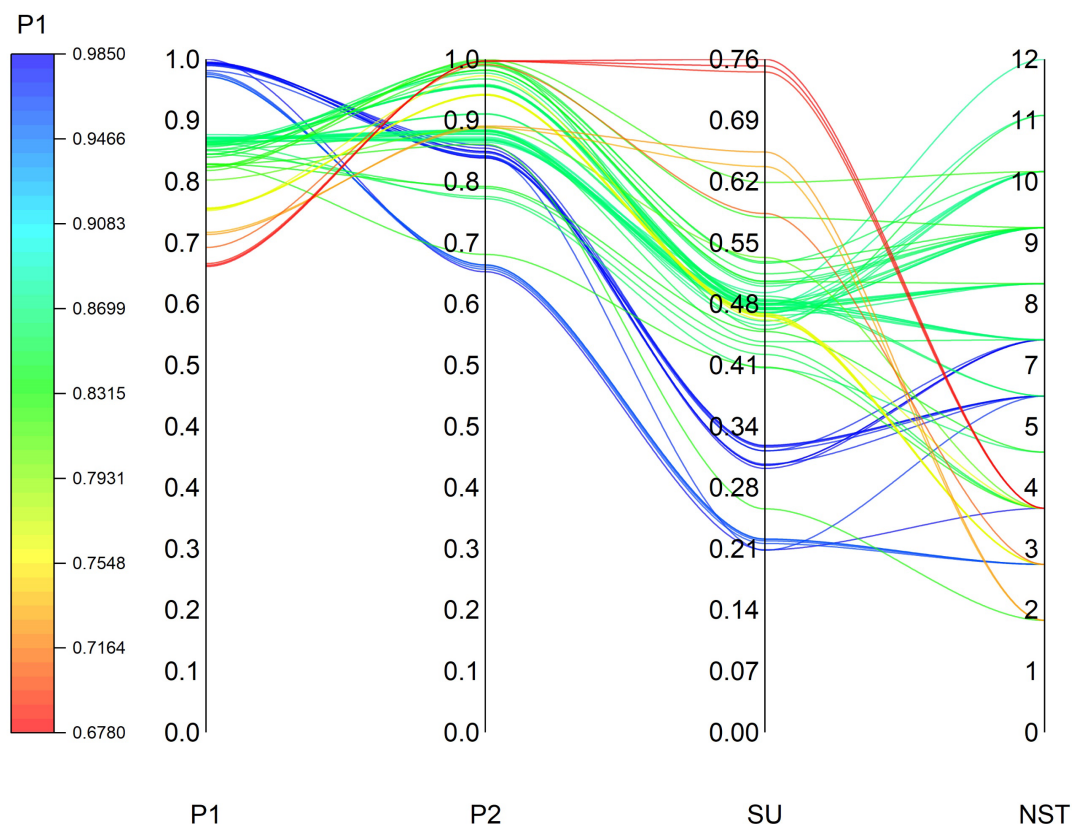


Figure 5. Parallel coordinate plot of the non-dominated solutions obtained by Fresa to solve the four-objective problem defined by Eqs. (6)-(9). P1 and P2 correspond to the purities of Nd and Sm, respectively, SU is the organic solvent usage in m^3/yr , and NST is the number of stages total.

aqueous mixtures are summarized in Table 2.

As discussed previously, a vector of design variables, $\mathbf{x} = [q_j, \xi_j, D]$, is employed to optimize the metal recovery process. These variables are bounded due to operational (e.g., flow rate ratio to maintain segmented flow) or structural (e.g., channel length) constraints. Although, channel length, L_j , and number of stages, n_j , are not directly leveraged, they are implicitly defined through the ξ_j variable; hence, upper and lower bounds for those design pseudo-variables are also required. Table 3 summarizes the lower and upper bounds for the physical and pseudo- design variables.

Table 2: Thermophysical properties of the liquid mixtures in the process flowsheet.

Property	Value	
	Organic	Aqueous
Interfacial tension, γ (N m^{-1})	0.01	0.01
Viscosity, μ ($\text{kg m}^{-1}\text{s}^{-1}$)	0.0023	0.001
Density, ρ (kg m^{-3})	844	997

Table 3: Lower and upper bounds for the design (pseudo-) variables. Regular variables appear in bold while pseudo-variables in plain font.

Variable	Value	
	Lower	Upper
q_j (–)	0.5	2.0
ξ_j (–)	0.0	1.0
L_j (m)	0.05	0.4
n_j (–)	0	10
D (m)	0.001	0.005

Solving the three-objective problem resulted in 57 non-dominated solutions in 10101s (≈ 168 min or ≈ 2.8 hrs) on a 48-core Intel® Xeon® w7-2495X @4800 MHz and 250 GB of memory. The non-dominated solutions are summarized in a parallel coordinate plot of Fig. 3. The solutions in Fig. 3 appear color-coded in a scale from low to high purity of Nd. Fig. 3 shows a clear trade-off between the purity of the two components, demonstrated by the intersection of lines between the purity of Nd, P1, and the

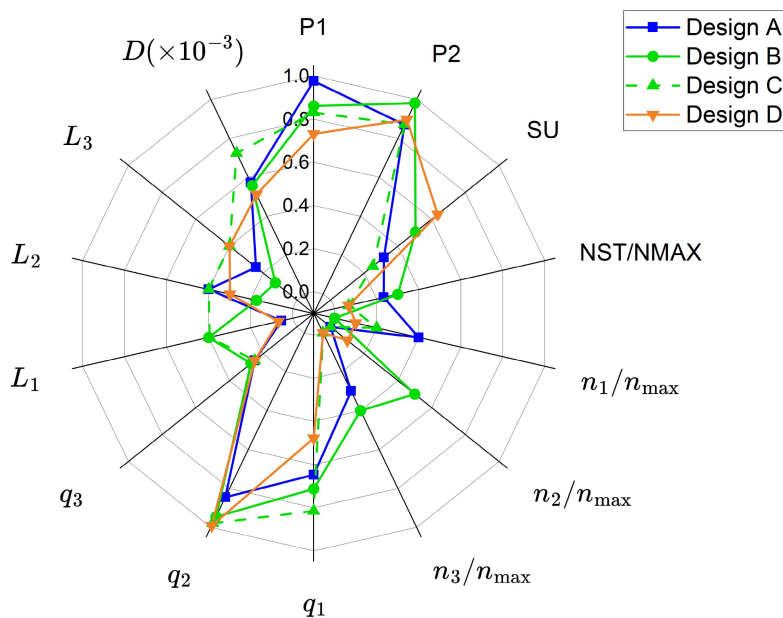


Figure 6. Radar chart of representative designs drawn from the set of non-dominated solutions (objectives and design variables) of the four-objective problem defined by Eqs. (6)-(9).

purity of Sm, P2. Therefore, the third objective, solvent usage, SU, is an important deciding factor, assuming that the first two objectives are of equivalent significance.

To gain better insight into the optimal designs, we have selected representative solutions to plot on the radar chart of Fig. 4, which shows the values of the objectives together with the values of the design variables. Design A results in a $P1 \approx 0.99$ and $P2 \approx 0.87$, whilst keeping $SU \approx 0.20$ requiring ten stages for E1 (the reader is referred to Fig. 1), no stages for E2, and four stages for E3. Employing E3, the scrubbing step, as well as not employing the second extraction step, E2, results in a very high purity of Nd at the expense of lowering the purity of Sm. Designs that had non-zero stages for E2 led to potentially lower P1, mainly due to back extraction and higher SU, since an additional stream of fresh organic solvent is employed, and thus were dominated by other solutions. Design B, on the other hand, is characterized by $P1 \approx 0.84$, $P2 \approx 0.99$, and $SU \approx 1.09$. While the trend in purities has reversed, vis-à-vis Design A, SU is significantly higher. Specifically, Design B disabled E1 (no stages), while it required five and nine stages for E2 and E3, respectively. However, enabling step E2 without enabling step E1 is essentially incoherent. If there are no stages for E1, no separation takes place, and the streams entering E3 are empty; hence, E3 stages in this design have no impact whatsoever in the objective values. The reason this design appeared in the set of non-dominated solutions is potentially due to the stochasticity of the optimization method that did not potentially converge to the global Pareto frontier, and although one would not normally

consider it, it still presents as a valid design. Another representative solution is that of Design C, which comprises of $P1 \approx 0.72$, $P2 \approx 0.99$, and $SU \approx 0.6$. Although this design achieves lower P1 compared to Design B, it achieves savings in SU by employing four stages for E1 and five stages for E2. To select an optimal design amongst the three representatives, one must weigh in process profitability (e.g., component purity vs price) and operational cost/environmental footprint (e.g., solvent cost). However, capital cost may be considered, as well. Although the total number of allowable stages is relatively small (i.e., 30) and might not impact considerably the economics of a pilot plant, the scaling out of the process, that is installing multiple channels in parallel for every stage to increase throughput, can have a big impact on capital cost. Therefore, we must investigate the minimization of total stages in addition to the three objectives we have already considered.

Note that although the superstructure does not consider the scaled-out process, capital cost is reflected through the number of stages per step. Thus, the fourth objective, in addition to the original three of Eqs. (6)-(8) reads:

$$\min_{\mathbf{x}} f_4(\mathbf{x}) \equiv \sum_{j=1}^3 n_j \quad (11)$$

The four-objective problem was solved using the same feeds, concentrations, and thermophysical properties as before, yielding 75 non-dominated solutions in 7136 s (≈ 118 min or ≈ 2 hrs). The resulting Pareto set, shown in Fig. 5, exhibits the same trade-off between Nd and Sm

purities observed in the three-objective case. Four representative designs are presented in Fig. 6. Design A achieves high purities ($P_1 \approx 0.98$, $P_2 \approx 0.87$) with moderate solvent usage ($SU \approx 0.32$) using seven total stages allocated to E1 and E3. Design B prioritizes Sm purity ($P_2 \approx 0.98$) but incurs higher solvent usage ($SU \approx 0.5$) and exhibits counterintuitive stage allocation, again disabling E1. Designs C and D highlight low-complexity alternatives: Design C achieves balanced purities with minimal solvent usage using only two stages in E1, while Design D trades Nd purity for a small gain in Sm purity at substantially higher solvent consumption.

The results show that no single optimal configuration exists; instead, families of non-dominated optimal designs emerge, reflecting trade-offs between separation performance, solvent usage, and total stages. While near-perfect Nd and Sm purities typically require multiple stages, comparable performance can often be achieved with fewer stages, albeit sometimes at higher solvent consumption. Overall, the multi-objective framework provides clear decision support for balancing performance, operational sustainability, and capital cost.

CONCLUSIONS

This work focused on modeling and multi-objective optimization to intensify the design of continuous flow LLE in small channels for critical metals recovery from e-waste. A superstructure-based approach was used to represent alternative flowsheet configurations with variable numbers of extraction and scrubbing stages, while mass transfer in segmented flow was described using simple differential balances to balance accuracy and computational efficiency. Discrete design decisions, such as stage allocation, were handled through a continuous pseudo-variable formulation to limit combinatorial complexity. The resulting nonlinear optimization problem was solved using the plant propagation algorithm implemented in Fresa.

The framework reveals key trade-offs between separation performance, solvent usage, and process complexity. Several non-dominated solutions show that high metal purities can be achieved with relatively simple configurations, highlighting opportunities for more compact and scalable designs. Overall, the approach supports informed selection of designs that balance performance with practical implementation constraints. Future work will consider tackling the intensification of metals recovery under feed variability conditions and parametric uncertainty.

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